Gauge Structure from Admissible Non-Injective Projection

Jérôme Beau
Independent Researcher
jerome.beau@cosmochrony.org

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Abstract

Starting from the axiomatic foundation of admissible non-injective transitions [1], we derive the structure of gauge theory without introducing it as a postulate. Gauge charges are identified as invariants of equivalence classes in the projection fibre: two configurations are gauge-equivalent when they project to the same observable state. The group of fibre-preserving transformations constitutes the gauge group. SU(2) is identified as the unique minimal non-abelian admissible group from the quaternionic structure of the admissible fibre; U(1) arises as the residual phase symmetry; SU(3) is a structurally motivated candidate extension whose derivation remains open. The gauge connection arises as the infinitesimal generator of fibre transport under varying projection, and Yang–Mills dynamics emerge from minimising the resulting curvature. No dynamical substrate is required at any step; the derivations rest solely on Axioms A1–A4 of [1] and the spectral results of the O-series. Effective spacetime geometry, requiring the continuum limit of [9], is left for a companion paper.

Keywords. gauge symmetry; non-injective projection; fibre bundles; gauge invariance; Yang–Mills theory; emergent gauge structure; admissibility; Heisenberg group; SU(2); quaternionic structure

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1 Introduction

Position in the programme

The companion papers [1, 2, 3] establish the quantum-mechanical structure of the Cosmochrony framework: the proto-state, phase coherence, the Born rule, the singlet correlator, and the Tsirelson bound, all derived from the admissibility structure of non-injective projection. These results concern the *internal* structure of the admissible fibre at a single transition step.

The present paper addresses one macroscopic consequence of allowing the projection structure to vary from point to point in the effective continuum: the conserved invariants of the projection fibre and the symmetry group they generate. The answer is gauge charges and gauge groups (§2), with Yang–Mills dynamics emerging from fibre transport (§4). Effective spacetime geometry, which depends on the continuum limit of [9], is left for a companion paper.

What is and is not derived in this paper

The present work establishes the structural origin of gauge symmetry and gauge connection from admissible non-injective projection. It is not intended as a full reconstruction of the Standard Model, but as an identification of the minimal structures that must emerge at the effective level.

This paper derives the *structure* of gauge theory from the admissibility framework. It does not derive specific coupling constants or the precise matter content of the Standard Model. These remain open and are identified as directions for future work in Section 5.

The derivations are structural: they identify which gauge groups are *admissible* given the constraints, not which specific values their parameters take.

Organisation

Section 2 derives gauge charges as fibre invariants, identifies the gauge group, and establishes SU(2) as the unique minimal non-abelian admissible group. Section 3 derives the gauge connection as the infinitesimal generator of fibre transport. Section 4 derives Yang–Mills dynamics from the curvature of this connection. Section 5 identifies the open directions.

2 Gauge Charges as Fibre Invariants

2.1 Charges from non-injectivity

The non-injectivity of Π_n (Axiom A2 of [1]) means that multiple configurations in $\Omega_n^{(c)}$ project to the same observable state O_n. These configurations are physically indistinguishable at the observable level but may carry distinct internal structure.

Definition 2.1 (Gauge charge)**.** *A* gauge charge *is a functional*

$$Q : \Omega_n^{(c)} \longrightarrow \mathbb{R} \tag{1}$$

that is constant on projection fibres:

$$\Pi_n(\omega_1) = \Pi_n(\omega_2) \implies Q(\omega_1) = Q(\omega_2). \tag{2}$$

A gauge charge is therefore an observable that cannot be resolved by Π_n itself — it lives in the kernel structure of Π_n — but is well-defined on the fibre.

Definition 2.2 (Gauge group)**.** *The* gauge group $\mathcal{G}$ *is the group of transformations $\omega \mapsto g \cdot \omega$ that preserve the projection:*

$$\mathcal{G} := \big\{ g \in \mathrm{Aut}(\Omega_n^{(c)}) \mid \Pi_n(g \cdot \omega) = \Pi_n(\omega) \text{ for all } \omega \big\}. \tag{3}$$

$\mathcal{G}$ consists precisely of the fibre-preserving symmetries: it acts within $\Pi_n^{-1}(O_n)$ without affecting the observable output.

Remark 2.3 (Gauge invariance as structural necessity)**.** *The gauge group is not introduced by hand. It is forced by the structure of Π_n: as soon as Π_n is non-injective, there exist non-trivial fibre-preserving transformations, and hence a non-trivial gauge group. Gauge invariance is a structural consequence of Axiom A2, not an additional postulate.*

2.2 Structural constraints on admissible gauge groups

Not every group is admissible under Definition 2.2. Three structural constraints restrict the possible gauge groups.

(i) *Non-factorisability.* By Axiom A3, the admissible fibre cannot decompose into independent components at each transition. This excludes purely abelian direct-product structures beyond a single residual phase factor.

(ii) *Compatibility with the admissible algebra.* The admissible fibre is identified with the Weil representation of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ [1], whose continuous limit admits a quaternionic minimal structure [3, 7]. By the classification of normed division algebras, the non-abelian part of $\mathcal{G}$ must embed into the group of unit quaternions, which is SU(2).

(iii) *Stability under admissibility refinement.* Under spectral refinement (large-q limit), admissible sectors must remain closed under the projection-induced transport. This favours compact Lie groups with stable representation theory.

These constraints do not uniquely determine the full gauge group, but they identify U(1) as the residual phase symmetry (constraint (i)), SU(2) as the minimal non-abelian admissible sector (constraint (ii)), and suggest SU(3) as a higher-dimensional extension compatible with fibre-preserving actions (constraint (iii), open).

2.3 Identification of the gauge groups

The three candidates are identified as follows.

- U(1): global phase rotations $\omega \mapsto e^{i\theta}\omega$ within the complex fibre $\mathbb{C}^q$. These preserve $|\omega|$ and hence the observable norm, while acting non-trivially on the phase — which is in the kernel of Π_n by the phase invariance of $\sigma_c(n)$ [6].
- SU(2): the stable fixed point of the admissibility flow established in [3] Theorem 8.1. Its fibre-preserving action on $\mathbb{C}^2$ is the spin-$\frac{1}{2}$ structure derived in [2]. Its identification as the *unique* minimal non-abelian admissible group is established as Theorem 2.5 below.
- SU(3): a structurally motivated candidate for a higher-dimensional internal symmetry of the fibre, corresponding to multiple co-admissible internal directions beyond SU(2). Its identification from the O-series admissibility data is an open direction (Section 5).

Remark 2.4 (Status of SU(3))**.** *The identification of* SU(3) *as a gauge group is a structural hypothesis, not a derived result. Establishing it would require finding, within the O-series spectral data, a co-admissibility pattern for a group with* SU(3) *structure analogous to the* $\lambda_{1/2} = \lambda_{3/2}$ *co-admissibility of [3]. This is identified as a medium-term open direction.*

2.4 Rigidity of the minimal non-abelian sector

Theorem 2.5 (Minimal admissible gauge sector)**.** *Under Axioms A1–A3 and Born–Infeld parity, the unique minimal non-abelian compact Lie group compatible with admissible non-injective projection is* SU(2)*.*

Structural argument. Axiom A3 (Born–Infeld admissibility) forbids factorisation of the fibre, enforcing a non-trivial commutator structure on the admissible algebra. By Foundation Theorem 5.6 [1], the admissible fibre F_n is

identified with the Weil representation V_ρ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. In the continuous admissible limit, the Born–Infeld parity involution and the quaternionic minimality result of O23 [7] force the image of any admissible morphism into $\mathrm{Im}\,\mathbb{H}$, the imaginary part of the quaternion algebra. This three-dimensional space is isomorphic to the Lie algebra $\mathfrak{su}(2)$. By the quaternionic rigidity established in O27 [8], any admissible morphism satisfying the three structural constraints of that paper necessarily factors through $\mathfrak{su}(2)$. Exponentiation yields SU(2) as the unique minimal non-abelian compact group. Among compact Lie groups, minimality forces a simple rank-one group; the only such group compatible with the quaternionic constraint is SU(2). □

Remark 2.6. *Theorem 2.5 is not a phenomenological selection:* SU(2) *is the* forced *output of the admissibility constraints, not a candidate chosen to match observations. The full details of the quaternionic rigidity argument are established in [7, 8].*

3 The Gauge Connection

3.1 The problem of comparing neighbouring fibres

The following development is conditional on two structural hypotheses. First, the admissible fibre F_n admits an effective continuum description in which it forms a smooth fibre bundle over an effective spacetime manifold. Second, this manifold itself carries a differentiable structure, which requires the emergence of a continuum geometry from the discrete admissibility filter — a step established only conditionally in [9] and not proved here. Neither hypothesis is an axiom of the present framework; both are working assumptions whose justification is deferred to [9] and a companion paper on effective geometry. All results in §3–4 are therefore conditional on these two hypotheses.

For each point x in the effective manifold, the fibre $F_x = \Pi^{-1}(x)$ carries the internal gauge structure of Section 2.

When the projection $\Pi(x)$ varies from point to point, the fibres F_x and F_{x+dx} are not canonically identified. A state $\psi(x)$ transforming under $\mathcal{G}$ acquires an additional term under differentiation:

$$\partial_\mu(g(x)\,\psi(x)) \;=\; g(x)\,\partial_\mu\psi(x) + (\partial_\mu g(x))\,\psi(x), \tag{4}$$

which is not covariant under local gauge transformations $g(x) \in \mathcal{G}$.

3.2 Connection as fibre transport operator

A comparison between fibres at neighbouring points x and $x + dx$ requires the specification of a transport map

$$T_{x\to x+dx} : F_x \;\longrightarrow\; F_{x+dx}. \tag{5}$$

Because the projection Π is non-injective, there is no canonical identification between fibres at distinct points. The transport map is therefore defined only up to a fibre-preserving transformation, i.e. an element of the gauge group $\mathcal{G}$.

The gauge connection arises as the infinitesimal generator of this transport:

$$T_{x \to x+dx} \;=\; \mathrm{Id} + A_\mu(x)\, dx^\mu + o(dx), \tag{6}$$

where $A_\mu(x)$ takes values in the Lie algebra $\mathfrak{g}$ of the gauge group. This definition does not rely on a background bundle structure: the connection is the minimal structure required to compare admissible fibres under variation of the effective state.

3.3 Gauge connection from fibre transport

Definition 3.1 (Gauge connection)**.** *A* gauge connection A_μ *is the* $\mathfrak{g}$*-valued one-form appearing in* (6)*, defining a covariant derivative*

$$D_\mu \;:=\; \partial_\mu + A_\mu \tag{7}$$

such that $D_\mu\psi$ *transforms homogeneously under local gauge transformations.*

Proposition 3.2 (Admissibility interpretation of A_μ)**.** *In the projection framework,* $A_\mu(x)$ *encodes the local variation of the admissible fibre structure: it is the infinitesimal correction needed to transport an admissible configuration from the fibre at* x *to the fibre at* $x + dx$ *consistently with the admissibility constraints.*

Structural argument. The admissibility constraints (A1–A3 of [1]) define, at each point x, a fibre F_x of structurally compatible configurations. When $\Pi(x)$ varies, the fibres F_x and F_{x+dx} are related by the differential of the projection map. Consistency of the admissibility condition across neighbouring fibres — i.e. that a configuration admissible at x maps to an admissible configuration at $x+dx$ — requires a linear correction of the form $A_\mu(x)\, dx^\mu$ taking values in $\mathfrak{g}$. This correction is precisely the gauge connection of Definition 3.1. □

Remark 3.3 (Relation to the holonomy argument)**.** *The white paper [5] establishes that non-injective admissible projections generically have non-trivial holonomy, which is the minimal structural condition for a gauge symmetry to arise as a descriptive invariance. Proposition 3.2 is the local infinitesimal version of this global statement.*

4 Yang–Mills Dynamics

4.1 Curvature as obstruction to fibre consistency

Given the transport map of §3.2, consider two infinitesimal displacements dx^μ and dx^ν from a point x. The failure of successive transports along these directions to commute defines the curvature of the connection.

Definition 4.1 (Field strength)**.** *The* field strength *associated with the gauge connection A_μ is the curvature of the covariant derivative:*

$$F_{\mu\nu} \;:=\; [D_\mu, D_\nu] \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \tag{8}$$

$F_{\mu\nu}$ measures the obstruction to consistent identification of admissible fibres over infinitesimal loops — equivalently, the non-commutativity of the admissibility constraints in two neighbouring directions. A flat connection ($F_{\mu\nu} = 0$) corresponds to a globally consistent fibre structure; non-zero curvature signals inconsistency in the transport of admissible configurations.

4.2 Yang–Mills action from admissibility minimisation

Structural origin of Yang–Mills-type dynamics. The admissibility principle (A3 of [1]) favours configurations that preserve the full multiplicity of open directions without premature selection. Translated to the gauge connection, this principle selects configurations that minimise the obstruction to consistent fibre transport, i.e. minimise the curvature $\|F_{\mu\nu}\|^2$.

The Cosmochrony-specific contribution is the identification of $F_{\mu\nu}$ as the natural curvature object: it follows from the non-injectivity of Π (via fibre transport, §3.2) without postulating a gauge field. The subsequent step — that the unique second-order gauge-invariant Lorentz-invariant functional of $F_{\mu\nu}$ is $\int \mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})$ — is a standard result of gauge theory, not specific to the present framework.

Remark 4.2 (Yang–Mills action)**.** *Under the continuum hypotheses of §3, the minimisation principle above selects the action*

$$S_{\mathrm{YM}}[A] \;=\; \int \mathrm{Tr}\big(F_{\mu\nu}F^{\mu\nu}\big)\,\sqrt{-g}\,d^4x, \tag{9}$$

whose variation yields the Yang–Mills equations $D_\mu F^{\mu\nu} = 0$. This is a structural motivation, not a derivation from Axioms A1–A4 alone. The uniqueness of (9) *follows from standard EFT power-counting (second order in A_μ, gauge-invariant, Lorentz-invariant); it does not require the Cosmochrony framework. What the framework contributes is the non-postulated origin of $F_{\mu\nu}$ itself. A direct derivation of* (9) *from the axioms via the Born–Infeld limit is discussed in [4]; derivation from Axioms A1–A4 alone remains open (Section 5).*

5 Open Directions

Direct derivation of Yang–Mills uniqueness. Remark 4.2 gives a structural motivation for the Yang–Mills action. Its uniqueness — that S_{YM} is the *only* admissibility-compatible second-order action — follows from the Born–Infeld limit [4] but not yet from Foundation M axioms alone.

SU(3) **from admissibility.** The identification of SU(3) as a gauge group requires finding a co-admissibility pattern for an SU(3)-type group in the O-series spectral data. The analogue of the $\lambda_{1/2} = \lambda_{3/2}$ result on $2I$ (established in [3]) must be found for a group with SU(3) structure.

Coupling constants. The present paper identifies which gauge groups are admissible. It does not determine the numerical values of coupling constants (g_s, g_w, α). These are expected to emerge from the quantitative spectral data of the O-series via the transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^* \approx 0.126$, but the explicit identification is open.

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